B.TECH 3RD SEM DISCRETE MATHEMATICS

BEU PATNA

SUBJECTIVE

QUESTION


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Q.1 Answer any SEVEN of the following (Objective/Short Answer): $[2 \times 7 = 14]$

(a) If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, find the symmetric difference $A \Delta B$.

(b) Define a Bijective function with a suitable example.

(c) State the Pigeonhole Principle. (PP)

(d) Find the Greatest Common Divisor (GCD) of 124 and 52 using the Euclidean Algorithm.

(e) Define a Tautology and a Contradiction in Propositional Logic.

(f) If a graph has 10 edges and every vertex has degree 2, how many vertices does the graph have?

(g) Define an Abelian group.

(h) State Cantor's Diagonal Argument.

(i) What is the Chromatic number of a Complete Graph K_n?

(j) Represent the statement "For every integer, there exists an integer y such that $x + y = 0$" using quantifiers.

Q.2

(a) Define an Equivalence Relation. Let R be a relation on the set of integers $\mathbb{Z}$ defined by xRy if and only if $(x - y)$ is divisible by 5. Prove that R is an equivalence relation. [7]

(b) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \frac{3x-4}{5}$. Show that f is a bijection and find its inverse f^{-1}. [7]

Q.3

(a) Find the GCD of 81 and 237 using the Euclidean Algorithm and express it in the form $81x + 237y$, where x and y are integers. [7]

(b) State and prove the Fundamental Theorem of Arithmetic. [7]

Q.4 ** 2024 PYQ

(a) Using the Principle of Mathematical Induction, prove that for all $n \geq 1$:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

[7]

(b) In a survey of 100 students, it was found that 50 play Football, 45 play Cricket, and 40 play Hockey. 20 play both Football and Cricket, 15 play Cricket and Hockey, and 10 play Football and Hockey. 5 students play all three games. Find the number of students who do not play any of the three games. [7]

Q.5

(a) Construct a truth table to verify the logical equivalence:

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$$

[7]

(b) Explain the Rules of Inference. Use them to show that the following premises lead to the conclusion r: $p \vee q$, $q \to r$, and $\neg p$. [7]

p	q	$\neg p$	$\neg q$	$\neg p \wedge q$	$\neg p \wedge \neg q$
F	F				
F	T				
T	F				
T	F				

Q.6

(a) Let $(G, \cdot)$ be a group. Prove that for any $a, b \in G, (a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$. [7]

(b) Define a Subgroup. Prove that a non-empty subset H of a group G is a subgroup if and only if for all $a, b \in H, ab^{-1} \in H$. [7]

Q.7

(a) Simplify the following Boolean expression using Boolean Algebra laws:

$$F(x, y, z) = x'yz + x'yz' + xy'z' + xy'z$$

[7]

Q.8

V + E - F = 2

(a) State and prove Euler's Formula for a connected planar graph with V vertices, E edges, and F faces. [7]

(b) Define Eulerian and Hamiltonian graphs. Provide an example of a graph that is Eulerian but not Hamiltonian. [7]

E

Q.9

(a) Prove by contradiction that $\sqrt{2}$ is an irrational number. [7]

(b) What is a Rooted Tree? Define the following terms with respect to trees: (i) Level of a node (ii) Height of a tree (iii) Leaf node (iv) Internal node. [7]